

Chapter 3

Scales and Direction

Class 10 - Total Geography Morning Star

Short Answer Questions

Question 1

How is direction shown on a map?

Answer

The direction is shown on the map with the help of North-South line . The arrow indicating up is considered as north and the arrow which indicates down, depicts south direction. The line perpendicular to the North South line represent the east-west direction. The line on the right depicts east direction and the line towards the left depicts the west direction.

Question 2

What do you mean by R.F.?

Answer

It is a method of representing scale on the map. It is expressed as a fraction showing the ratio of a unit distance on the map and the distance measured in the same units on the ground.

Question 3

What is the advantage of R.F.?

Answer

The main advantage of R.F. is that it is only a fraction and is independent of any particular unit of measurement. It can be converted into any particular unit of measurement and has universal application.

Question 4

Explain Magnetic declination with the help of a diagram.

Answer

The angle between the true north-south line and the magnetic north-south line is known as magnetic declination.

Question 5

Why are 'True North' and 'Grid North' different?

Answer

'True North' and 'Grid North' are different because of the spherical shape of the earth.

Question 6

Convert the following numerical scale (R.F.) into statement scales:

- (a) 1:1000
- (b) 1:50,000
- (c) 1:5,00,000

Answer

(a) R.F. = 1:1000 = 1 cm to 1000 cm
(Since 100 cm = 1 m, therefore 1000 cm = 10 m)
Hence, the scale is 1 cm to 10 m.

(b) R.F. = 1:50,000 = 1 cm to 50,000 cm
(Since 100 cm = 1 m, therefore 50,000 cm = 500 m)
Hence, the scale is 1 cm to 500 m.

(c) R.F. = 1:5,00,000 = 1 cm to 5,00,000 cm
(Since 1,00,000 cm = 1 km, therefore 5,00,000 cm = 5 km)
Hence, the scale is 1 cm to 5 km.

Question 7

Convert the following into numerical scales (R.F.):

- (a) 10 centimetres to kilometre
- (b) 1 centimetre to 20 kilometres
- (c) 2 cm to 5 km
- (d) 6 inch to 1 mile
- (e) 1 inch to 2 1/2 miles
- (f) 7 cm to 6.3 km

Answer

(a) The scale is 10 cm to 1 km = 10 cm to 1,00,000 cm
(Since 1 km = 1,00,000 cm)

R.F. = Distance on the map / Distance on the ground

R.F. = 10 / 1,00,000

R.F. = 1:10,000

(b) The scale is 1 cm to 20 km = 1 cm to 20,00,000 cm
(Since 1 km = 1,00,000 cm)

R.F. = Distance on the map / Distance on the ground

R.F. = 1 / 20,00,000

R.F. = 1:20,00,000

(c) The scale is 2 cm to 5 km = 2 cm to 5,00,000 cm
(Since 1 km = 1,00,000 cm)

R.F. = Distance on the map / Distance on the ground

R.F. = 2 / 5,00,000

R.F. = 1:2,50,000

(d) The scale is 6 inch to 1 mile = 6 inch to 63,360 inches
(Since 1 mile = 63,360 inches)

R.F. = Distance on the map / Distance on the ground

R.F. = 6 / 63,360

R.F. = 1:10,560

(e) The scale is 1 inch to 2.5 miles = 1 inch to 1,58,400 inches
(Since 1 mile = 63,360 inches)

R.F. = Distance on the map / Distance on the ground

R.F. = 1 / 1,58,400 R.F. = 1:1,58,400

(f) The scale is 7 cm to 6.3 km = 7 cm to 6,30,000 cm
(Since 1 km = 1,00,000 cm)

R.F. = Distance on the map / Distance on the ground

R.F. = 7 / 6,30,000

R.F. = 1:90,000

Distinguish between

Question 1

Statement Scale and Graphic Scale

Answer

Statement Scale	Graphic Scale
Scale is stated in words. We make a statement about it.	Scale is represented by a straight line divided into equal parts. These markings represent on the actual ground.
If the map is reduced or enlarged, a new scale needs to be worked out.	This scale stands valid even when the map is reduced or enlarged.

Question 2

True North and Magnetic North

Answer

True North	Magnetic North
True north is the direction indicated by the north star.	The north to which the compass needle points is called magnetic north.
True north is a fixed point on the globe.	Magnetic north is not a fixed point.

Answer the following questions

Question 1(a)

What is a Scale? Name the main methods of representing the scale of a map.

Answer

The scale of a map denotes the proportion that the distance between any two points on the map bears to the distance between the same two points on the surface of the earth.

The main methods of representing the scale of a map are-

1. A statement
2. Graphic or Linear scale
3. Representative Fraction

Question 1(b)

Convert the following statements into R.F.

- (a) 25 cm on the map = 5 km on ground.
- (b) 2½ inches on the map = 5 miles on ground.
- (c) 7 cm on the map = 63000 metres on ground.
- (d) 5 cm = 500 metres.
- (e) 15 cm = 6 km.
- (f) 10 cm = 1 km.
- (g) 12 cm = 72000 metres.

Answer

(a) The scale is 25 cm to 5 km = 25 cm to 5,00,000 cm
 (Since 1 km = 1,00,000 cm)
 $R.F. = \text{Distance on the map} / \text{Distance on the ground}$
 $R.F. = 25 / 5,00,000$
 $R.F. = 1:20,000$

(b) The scale is 2½ inches to 5 miles = 2.5 inches to 3,16,800 inches
 (Since 1 mile = 63,360 inches)
 $R.F. = \text{Distance on the map} / \text{Distance on the ground}$
 $R.F. = 2.5 / 3,16,800$
 $R.F. = 1:1,58,400$

(c) The scale is 7 cm to 63000 m = 7 cm to 63,00,000 cm
 (Since 1 m = 100 cm)
 $R.F. = \text{Distance on the map} / \text{Distance on the ground}$
 $R.F. = 7 / 63,00,000$
 $R.F. = 1:9,00,000$

(d) The scale is 5 cm to 500 m = 5 cm to 50,000 cm
 (Since 1 m = 100 cm)
 $R.F. = \text{Distance on the map} / \text{Distance on the ground}$
 $R.F. = 5 / 50,000$
 $R.F. = 1:10,000$

(e) The scale is 15 cm to 6 km = 15 cm to 6,00,000 cm
 (Since 1 km = 1,00,000 cm)
 $R.F. = \text{Distance on the map} / \text{Distance on the ground}$
 $R.F. = 15 / 6,00,000$
 $R.F. = 1:40,000$

(f) The scale is 10 cm to 1 km = 10 cm to 1,00,000 cm
 (Since 1 km = 1,00,000 cm)
 $R.F. = \text{Distance on the map} / \text{Distance on the ground}$
 $R.F. = 10 / 1,00,000$
 $R.F. = 1:10,000$

(g) The scale is 12 cm to 72000 m = 12 cm to 72,00,000 cm
 (Since 1 m = 100 cm)

R.F. = Distance on the map / Distance on the ground
R.F. = $12 / 72,00,000$
R.F. = 1:6,00,000

Question 2

Convert the following representative fractions into statements:

- (i) 1: 63360 (to show miles).
- (ii) 1: 1000000 (to show kilometres).
- (iii) 1: 100 (to show metres).
- (iv) 1: 10 (to show metres).
- (v) 1: 200,000 (to show kilometres).
- (vi) 1: 50 (to show metres and centimetres).

Answer

(i) R.F. = 1:63360
That means 1 inch on the map = 63360 inches on the ground
The scale is 1 inch to 1 mile
(Since 1 mile = 63,360 inches)

(ii) R.F. = 1:10,00,000
That means 1 cm on the map = 10,00,000 cm on the ground
The scale is 1 cm to 10 km
(Since 1 km = 1,00,000 cm)

(iii) R.F. = 1:100
That means 1 cm on the map = 100 cm on the ground
The scale is 1 cm to 1 m
(Since 100 cm = 1 m)

(iv) R.F. = 1:10
That means 1 cm on the map = 10 cm on the ground
The scale is 10 cm to 1 m
(Since 1 m = 100 cm)

(v) R.F. = 1:2,00,000
That means 1 cm on the map = 2,00,000 cm on the ground
The scale is 1 cm to 2 km
(Since 1 km = 1,00,000 cm)

(vi) R.F. = 1:50
That means 1 cm on the map = 50 cm on the ground
The scale is 10 cm to 5 m
(Since 1 m = 100 cm)

Question 4

The distance between New Delhi Station and Safdarjung Enclave bus stop is 20 km. On the map of Delhi, it has been shown by a line of 3.5 cm. Draw the linear scale of the map and calculate the R.F.

Answer

According to the given question,

The scale is 3.5 cm to 20 km = 3.5 cm to 20,00,000 cm

(Since 1 km = 1,00,000 cm)

R.F. = Distance on the map / Distance on the ground

R.F. = 3.5 / 20,00,000

R.F. = 1:571428.57

Question 5

On the map of India the cartographer forgot to draw the scale of the map. The student who knows the distance between Meerut and Delhi (70km), was asked to complete the scale. How will the student draw the scale if he measures the distance between Delhi and Meerut on the map to be 5 cm? Give the procedure, draw the scale and find out the R.F.

Answer

According to the given question,

The scale is 5 cm to 70 km = 5 cm to 70,00,000 cm

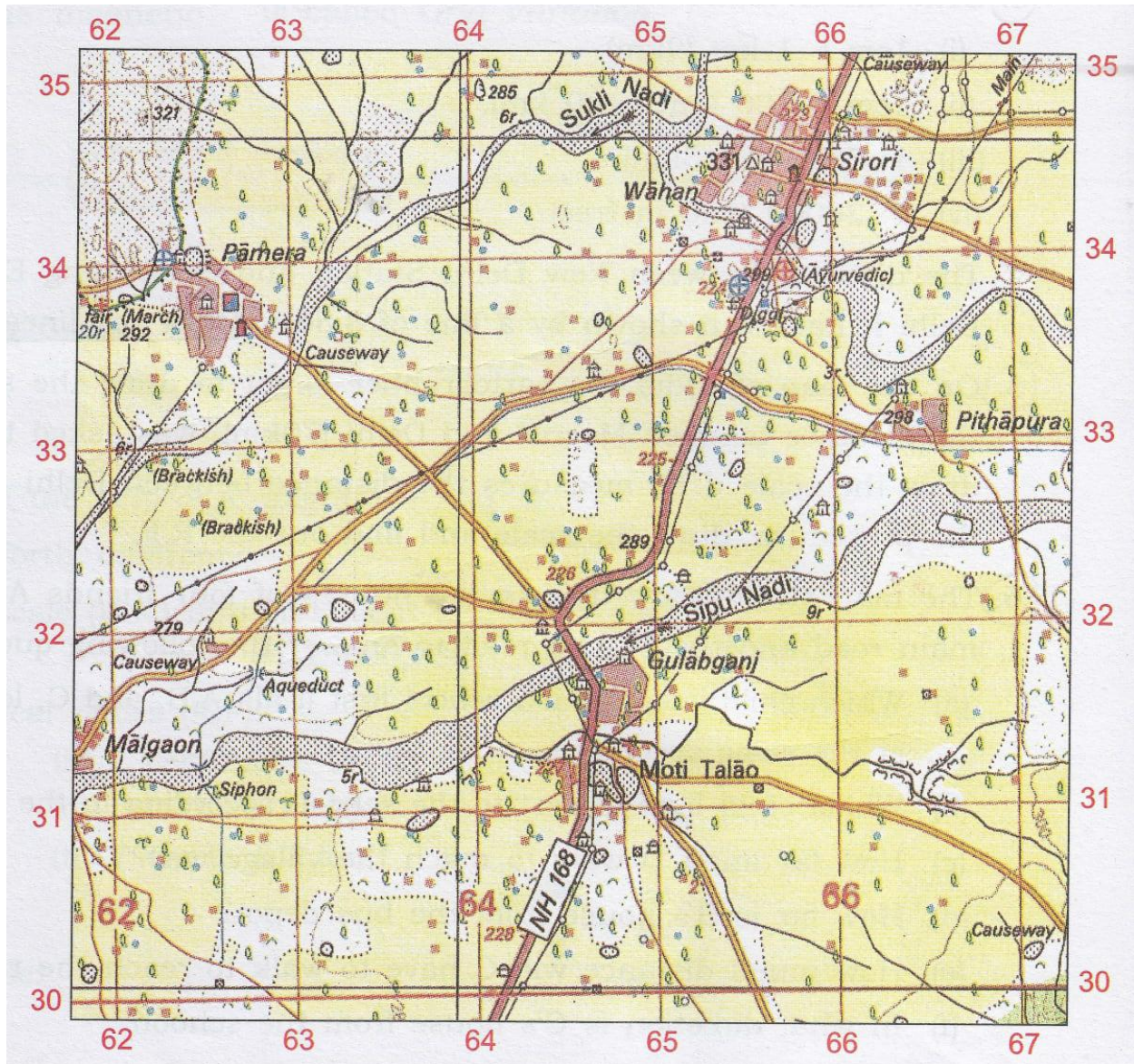
(Since 1 km = 1,00,000 cm)

R.F. = Distance on the map / Distance on the ground

R.F. = 5 / 70,00,000

R.F. = 1:14,00,000

Study the given map extract and answer the following questions



Question 1

Find the area in sq. km. of the area enclosed within the Eastings 63 to 66 and Northings 30 to 33.

Answer

Number of complete squares = 9

Since each grid square measures 1 km x 1 km, the area of the area enclosed within the Eastings 63 to 66 and Northings 30 to 33 will be 9 sq. km.

Question 2

Find the compass direction of the following:

- (i) Gulabganj from Pamera
- (ii) Malgaon from Sirori

Answer

- (i) Gulabganj lies to the south-east of Pamera.
- (ii) Malgaon lies to the south-west of Sirori.